

Patient Management Portal - Solution Design v2.1

Confidential - Internal Use Only

Version 1.0

1. Executive Summary

This document describes a HIPAA-compliant patient management portal for Greenfield Health Network.

The portal serves 50,000+ patients across 12 clinic locations with 200+ healthcare providers. Core capabilities: patient records, appointment scheduling, telehealth, prescriptions, and billing.

2. System Architecture

Cloud-native deployment on Azure with HIPAA BAA compliance.

Frontend: React.js with Material UI, WCAG 2.1 AA accessible.

Backend: .NET 8 Web API with Entity Framework Core.

Database: Azure SQL with Transparent Data Encryption and geo-replication.

Integration: HL7 FHIR R4 for interoperability with EHR systems.

3. Patient Authentication

REQ-HCP-001: Patients must authenticate using MFA via SMS OTP or authenticator app.

REQ-HCP-002: Biometric login (FaceID/TouchID) must be supported on mobile devices.

REQ-HCP-003: Session timeout after 15 minutes of inactivity with secure re-authentication.

REQ-HCP-004: Password policy requires minimum 12 characters with complexity rules.

REQ-HCP-005: Account lockout after 5 failed attempts with email notification to patient.

4. Patient Records Management

REQ-HCP-006: Patient profile must display demographics, allergies, medications, and conditions.

REQ-HCP-007: Medical history timeline with chronological view of visits, tests, and diagnoses.

REQ-HCP-008: Lab results must be available within 2 hours of processing by the lab system.

REQ-HCP-009: Patients can download their complete medical record in CCD-A format.

REQ-HCP-010: Document upload for insurance cards, referral letters, and prior authorizations.

5. Appointment Scheduling

REQ-HCP-011: Calendar view showing available slots by provider, specialty, and location.

REQ-HCP-012: Online booking with real-time availability checking against provider schedules.

REQ-HCP-013: Automated appointment reminders via SMS and email at 48hr and 2hr before.

REQ-HCP-014: Cancellation and rescheduling must be available up to 4 hours before appointment.

REQ-HCP-015: Waitlist functionality to fill cancelled slots automatically.

6. Provider Communication

REQ-HCP-016: Secure messaging between patients and providers with end-to-end encryption.

REQ-HCP-017: Message response SLA tracking with 24-hour target for non-urgent messages.

REQ-HCP-018: Attachment support for images, documents, and lab results in messages.